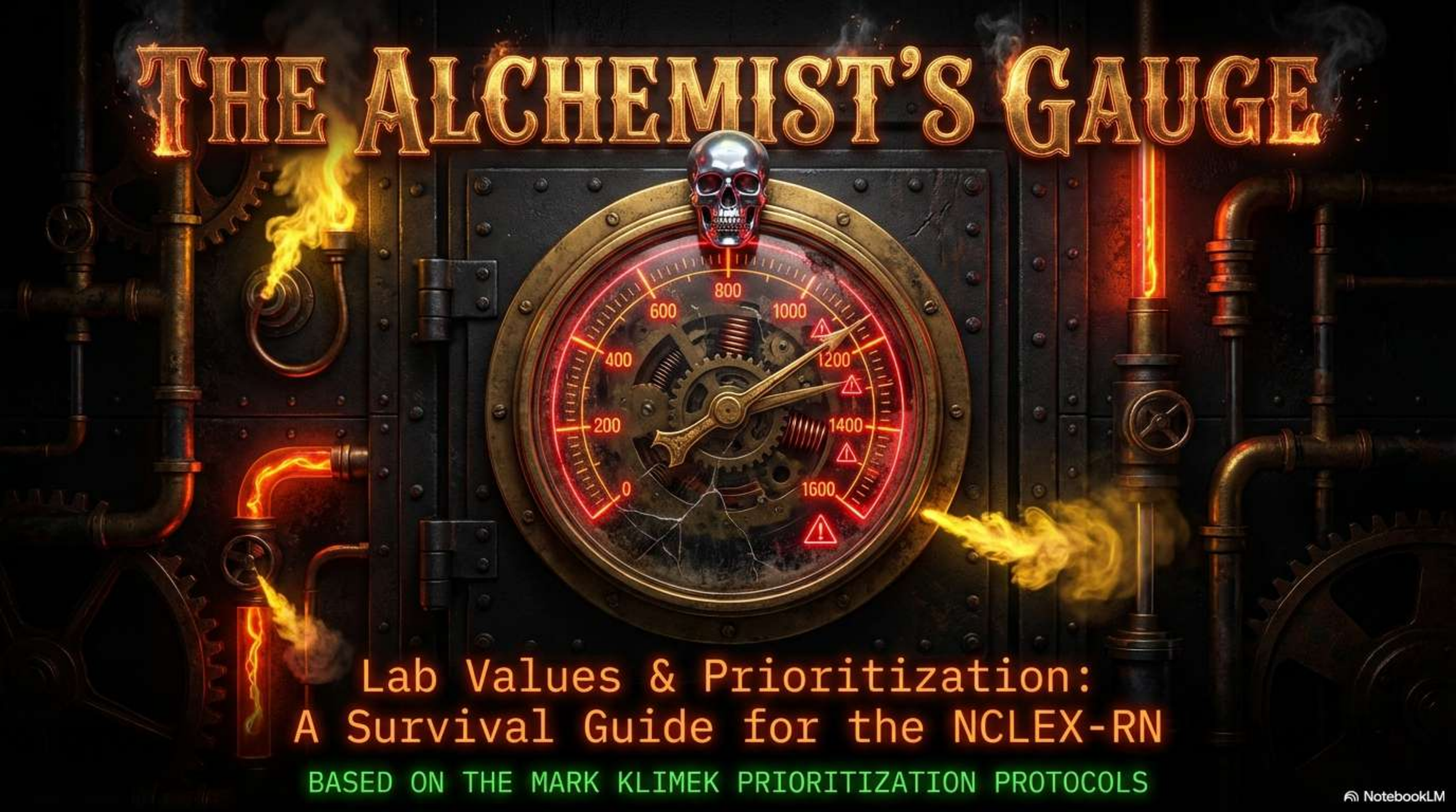
The Alchemist's Gauge



Lab Values & Prioritization: A Survival Guide for the NCLEX-RN

BASED ON THE MARK KLIMEK PRIORITIZATION PROTOCOLS

NotebookLM

CRITICAL INTERVENTION PROTOCOL: SURVIVAL MANUAL

LEVEL A

Low Priority.
Abnormal but Safe.
DO NOTHING.

LEVEL B

Monitor.
Abnormal & Concerned.
WATCH CLOSELY.

LEVEL C

Critical.
High Priority.
DO SOMETHING NOW.

LEVEL D

Deadly.
Highest Priority.
DO NOT LEAVE BEDSIDE.

1. HOLD → 2. ASSESS → 3. PREPARE → 4. CALL

NotebookLM

CREATININE: 0.6 - 1.2 mg/dL

Best indicator of kidney function.
Priority Level A.

- Abnormal = Kidney Disease (Not dying today).
- Exception: Dye Procedures (Contrast) require Morning Morning Call.

BUN: 8 - 25

Nitrogen Waste.

Priority Level B.

- Bun = 8 in a pack.
- Elevated = Check for DEHYDRATION.

NotebookLM

THE THINNER: INR
(Coumadin/Warfarin)

Normal Therapeutic Range: 2 - 3

CRITICAL VALUE: INR > 4 = LEVEL C

1. HOLD Warfarin
2. ASSESS for Bleeding
3. PREPARE Vitamin K
4. CALL Doctor

NotebookLM

THE PUMP ENERGY: POTASSIUM (K+)

Normal: 3.5 - 5.3 mEq/L

LOW (< 3.5)	HIGH (5.4 - 5.9)
Assess Heart	Hold K+
Prepare K+	Assess Heart
Call MD	Prepare Kayexalate / D5W / Insulin
WARNING: Never IV Push!	Call MD

NotebookLM

POTASSIUM ≥ 6.0: LEVEL D (DEADLY)
IMMINENT CARDIAC ARREST. DO NOT LEAVE BEDSIDE.
NURSE 1: Stay with patient. Hold Meds. Assess EKG.
NURSE 2: CALL STAT. Prepare D5W + Regular Insulin + Kayexalate.
Mnemonic: Sodium Swells, Potassium Pumps.
NotebookLM

THE ACID BATH: pH

Normal pH: 7.35 - 7.45

CRITICAL VALUE:
pH in the 6s
(e.g., 6.8) = LEVEL D

As pH goes down, the patient goes down. System Shutdown.

PROTOCOL:
1. ASSESS VITALS (Check for life).
2. CALL MD STAT.

Note: Only the MD can treat the underlying cause. Nurse supports life.

NotebookLM

THE EXHAUST FUMES: CO_2

Share Tech Mono

Normal: 35 - 45

HIGH (50s) = LEVEL C

Share Tech Mono

Sign: Discomfort/Dyspnea
Action: PURSED LIP BREATHING (Blow out the candle).
Warning: DO NOT give O2.

EXTREME (60s) = LEVEL D

Share Tech Mono

Sign: Respiratory Failure
Action: PREPARE TO INTUBATE & VENTILATE.
Call Respiratory Therapy + MD.

NotebookLM

THE INTAKE VALVE: pO2 (Arterial Blood Gas)

Normal: 78 – 100

LOW (70 - 77) = LEVEL C

Action: GIVE OXYGEN.

Outcome: Dyspnea usually resolves.

HYPOXIC FAILURE (60s) = LEVEL D

Action: INTUBATE & VENTILATE. Call Respiratory Therapy + MD.

RESPIRATORY FAILURE CRITERIA:
$CO_2 \geq 60$ AND $pO_2 \leq 60$

NotebookLM

THE LIFE FLUID: HEMATOLOGY

HEMOGLOBIN (Hgb)

- Normal: 12 - 18
- 8 - 11: Level B (Monitor Bleeding/Anemia)
- < 8: Level C (TRANSFUSE)

HEMATOCRIT (Hct)

- Normal: 36 - 54 (3x Hgb)
- Elevated: Level B (Assessment: DEHYDRATION)

NotebookLM

THE DEFENDERS: IMMUNITY

LEVEL C THRESHOLDS:

- TOTAL WBC < 5,000
- ANC (ABSOLUTE NEUTROPHIL COUNT) < 500
- CD4 COUNT < 200 (AIDS)

NEUTROPENIC PRECAUTIONS

- STRICT HANDWASHING
- NO FRESH FLOWERS / POTTED PLANTS
- NO RAW FRUIT / VEGGIES
- NO STANDING WATER (PITCHERS > 15 MINS)
- DEDICATED EQUIPMENT

NotebookLM

THE PATCH: PLATELETS

Normal: 150k - 400k

⚠ < 90,000 = LEVEL C ⚠

- Action: Bleeding Precautions (No IM meds, soft toothbrush, fall risk).

⚠ < 40,000 = LEVEL D ☠

- Risk: Spontaneous Hemorrhage.
- Action: TRANSFUSE.

NotebookLM

THE SALT & THE STRETCH

SODIUM (135 - 145)

- High = dEhydration
- Low = Overload
- CRITICAL EXCEPTION: If Change in LOC = LEVEL C (Safety Issue).

BNP (< 100)

- Elevated = Level B
- Indicator of CHF (Congestive Heart Failure).
- Note: Chronic Condition. Monitor, do not panic.

NotebookLM

THE 5 DEADLY 'D's

Highest Priority. DO NOT LEAVE THE BEDSIDE.

⚠️ 1. pH in the 6s (Severe Acidosis) ⚠️

☠️ 2. POTASSIUM in the 6s (Cardiac Arrest) ☠️

☠️ 3. CO2 in the 6s (Respiratory Failure) ⚠️

⚠️ 4. pO2 in the 6s (Hypoxic Failure) ⚠️

⚠️ 5. PLATELETS < 40,000 (Spontaneous Hemorrhage) ⚠️

NotebookLM

THE ALCHEMIST'S PHARMACOLOGY

1 ACID-BASE RULE

R.O.M.E. = Respiratory Opposite, Metabolic Equal.

2 TREATING HIGH POTASSIUM

- Insulin + D5W (Drives K+ into cell).
- Kayexalate (Sodium Polystyrene) (Removes K+ via body).

3 QUICK FIXES

- Anxiety (Resp Alkalosis) -> Paper Bag (Re-breathe CO_2).
- Opioid OD (Resp Acidosis) -> Narcan (Naloxone).

NotebookLM